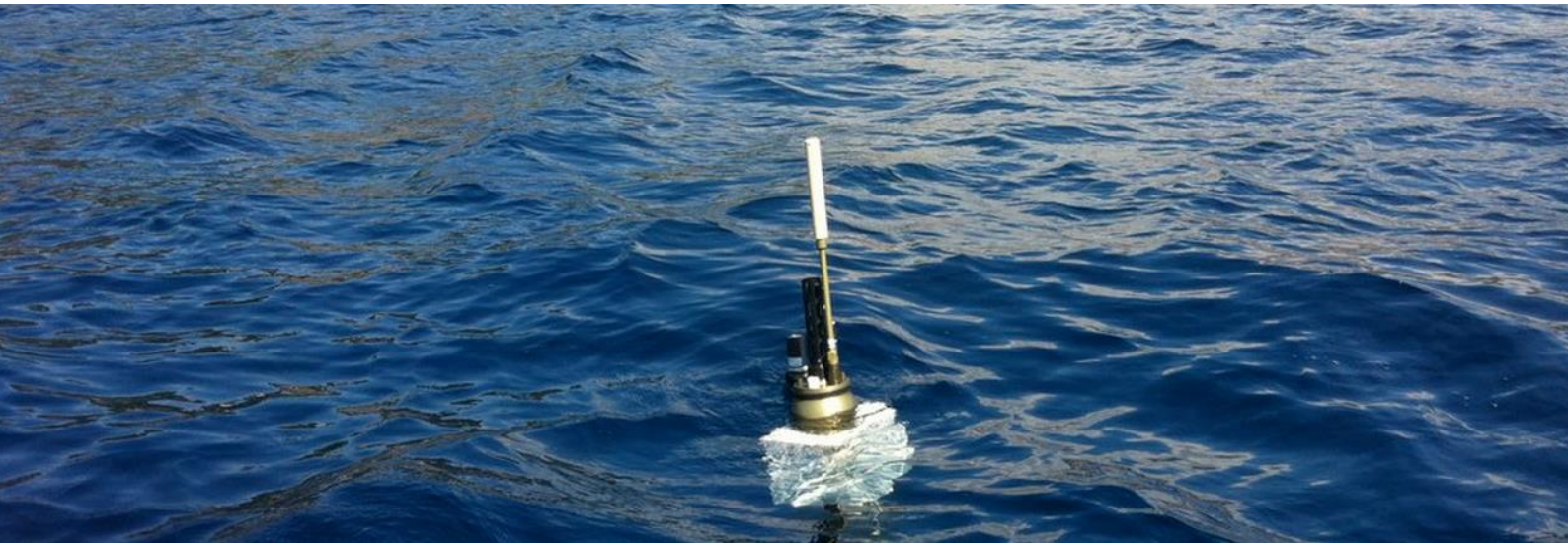
PFV2 Profiler - File management

PROFILING **F**LOAT **V**ECTOR **2**



NTxxxxxx_PFV2_File_Management_rev3.docx
Revision 1.2 (2024-09-17)

Table of contents

Table of contents 2
1. Revision history 3
1.1 Firmware upgrade 4
2. File system 5
2.1. Memory organisation 5
2.2. File types 5
2.3. Main memory – Working memory 5
2.4. Extended memory – Storage memory 5
2.5. GZ compression 6
2.6. SBD decoding 6
3. Configuration files 0
3.1 Local configuration via WiFi 0
3.2 Remote configuration via remote commands 0
3.3 Transmission of the configuration file 0
4. Technical files 2
4.1. Transmission of technical information 2
4.1.1. Self-test files 2
4.1.2. Mission files 2
4.1.3. End-of-life files 2
4.2. Technical information available 3
4.2.1. Product information events 3
4.2.2. Phases events 3
4.2.3. Security events 6
4.2.4. Alarms events 7
4.2.5. Float infos events 8
4.2.6. Sensors state events 9
4.2.7. Selftest events 10
4.2.8. Transmission events 12
4.2.9. Navigation events 13
4.2.10. Treatment events 13
5. Sensor data files 17

1. Revision history

Revision	Release date	Notes	Author
1.0	2024-01	Original	D. NOGRE
1.1	2024-07-03	Update (addons in yellow color)	D. NOGRE
1.2	2024-07-11	Add new technical messages “1022” & “4006”	D. NOGRE

1.1 Firmware upgrade

The following are the main changes to embedded software:

PFV2	Notes

2. File system

Float management is based on a specific file system developed by nke. This file system allows information and data to be stored when the float is turned off, together with the mission configuration.

These files can be accessed by an SFTP client via the WiFi connection.

2.1.Memory organisation

The memory space is divided into different areas intended for different uses. In particular, the following spaces can be found:

- Reserved: restricted system used to store file manipulation and management information
- Working: used for temporary storage (awaiting transmission).
- Storage: used to expand the temporary storage capacity of the workspace.

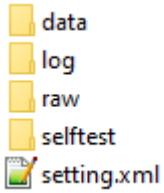
2.2.File types

The system uses various file types, as needed:

- Configuration file for mission and sensor settings
- Sensor data files (raw, processed and transmitted data)
- Technical data files
- Execution trace files (mission progress monitoring)
- Specific files (restricted system use)

2.3.Main memory – Working memory

The working memory (fast access) is itself divided into 4 directories:



- Folder for storing the current processed data
- Folder for storing the execution trace files
- Folder for storing the current raw data
- Folder for storing the result of the last selftest
- Configuration file embedding the mission and sensors descriptions

2.4.Extended memory – Storage memory

The use of an additional memory (slow access) increases the system's storage. In this case, files awaiting of transmission and to be moved from the working memory to the extended memory.

archive	-	Folder for storing the data already transmitted
data	-	Folder for storing the data in waiting of transmission
raw	-	Folder for storing the raw data of all previous missions and cycles

2.5.GZ compression

GZ compression is applied to all files. This compression is an option that can be deactivated if necessary. However, it is recommended that it is retained to improve transmission performance. The fact that there is less data to transmit saves energy and reduces Iridium's costs.

The principle of GZ compression involves a single file per compression. This file is then identified by keeping its name and adding the ".gz" extension.

If one of these files is fragmented, the cut can be made at any point in the file. The name extension is then formatted as ".gz.xxx", where "xxx" is the file assembly number.

To keep the filename size as small as possible, the file is renamed during compression. This name is replaced by a unique hexadecimal transmission number. Each file will therefore take a different number which will be incremented over transmissions. An optimization only possible since the original name of the file is integrated into the compression.



2.6.SBD decoding

Iridium SBD transmission is based on sending a message with a maximum size of 360 bytes (9602/9603 modem). If a larger file is sent, fragmentation is necessary. To enable the file to be reconstituted after transmission, a header is included at the beginning of this buffer for each transmission.

2.6.1. Header

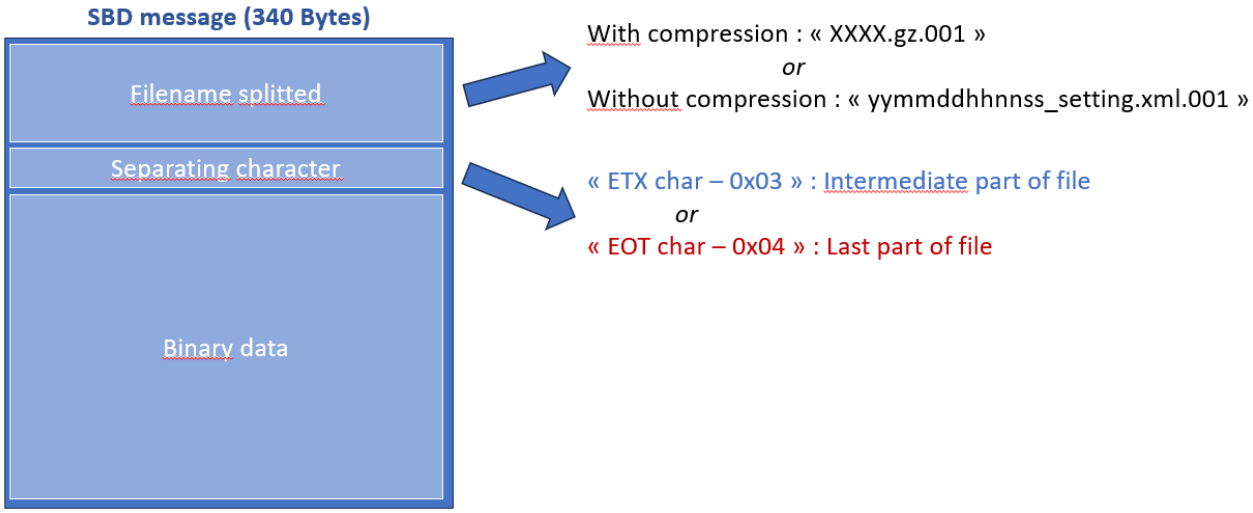
This header contains the name of the fragmented file and a separator (specific character). The purpose of the separator is to define the boundary between the header and the file's data, and to identify the last SBD of the fragmented file.

2.6.2. Separating character

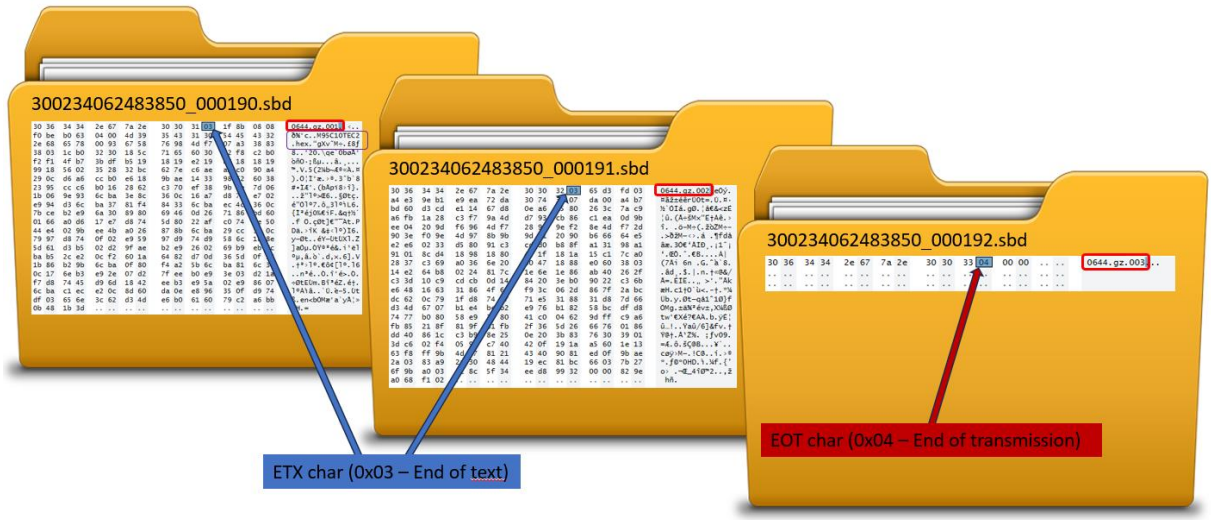
The character can only take 2 values:

- "ETX" char (0x03) meaning that it is an intermediate part of the file
- "EOT" char (0x04) meaning this is the last part of the fragmented file

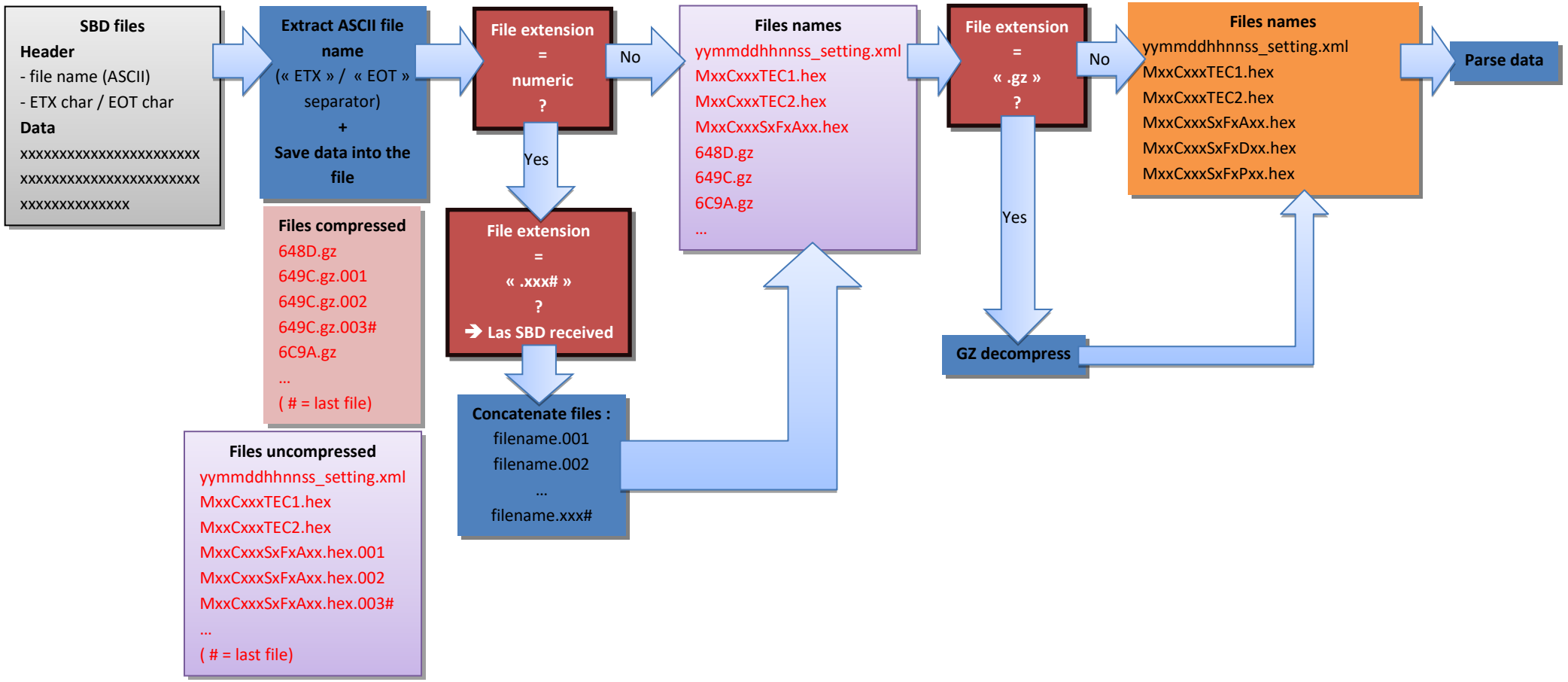
SBD content:



SBD encoding illustration:



2.6.3. Principle



P.S.: Python routines available, please contact nke Instrumentation.

3. Configuration files

The mission configuration is saved as a XML file organised by nodes and tags.

3.1 Local configuration via WiFi

During the mission configuration phase performed locally via WiFi, this file can be modified by the user. It can be manually uploaded in full using a SFTP client or partially modified with a command interpreter via a SSH client.

3.2 Remote configuration via remote commands

When the mission is in progress, the file can be modified by sending remote commands via the satellite communication system (SBD, RUDICS...).

3.3 Transmission of the configuration file

The system transmits the current configuration file at various stages of the mission:

- At the start of the mission (initial configuration)
- Upon a change of configuration following the application of a remote command (optional for an Iridium SBD transmission)

The various files generated are named using the format:

Iridium SBD : "yymmddhhnnss_SBD_cmd.ini"

- with "yymmddhhnnss": Date/Time of the float (12 characters)

Iridium RUDICS : “yymmddhhnnss_RUDICS_cmd.txt”

- with "yymmddhhnnss": Date/Time of the float (12 characters)

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<SETTING P0="347F8EFF">
<PRODUCT P0="ARVOR" P1="IRIDIUM SBD"/>
<MISSION P0="0">
        <PRESSURE_DETECTION P0="300" P1="10" P2="90"/>
        <INITIALIZATION P0="0"/>
        <LOOP01 P0="1">
                <CYCLE01 P0="1" P1="0" P2="">
                        <TRANSMISSION01 P0="false"/>
                        <BUOYANCY_RED P0="1"/>
                        <DESC_PARK P0="1000" P1="2.5"/>
                        <PARK_DRIFT P0="0" P1="50" P2="30"/>
                        <DESC_PROF P0="2000" P1="2.5"/>
                        <PROF_DRIFT P0="0" P1="50" P2="30"/>
                        <ASCENT P0="0" P1="9"/>
                        <EMERGENCE/>
                        <IN_AIR P0="0" P1="300"/>
                        <TRANSMISSION02/>
                        <END_OF_CYCLE P0="0"/>
                </CYCLE01>
                <CYCLE02 P0="999" P1="864000" P2="12:00:00">
                        <TRANSMISSION01 P0="true"/>
                        <BUOYANCY_RED P0="1"/>
                        <DESC_PARK P0="1000" P1="2.5"/>
                        <PARK_DRIFT P0="0" P1="50" P2="30"/>
                        <DESC_PROF P0="2000" P1="2.5"/>
                        <PROF_DRIFT P0="0" P1="50" P2="30"/>
                        <ASCENT P0="0" P1="9"/>
                        <EMERGENCE/>
                        <IN_AIR P0="2" P1="300"/>
                        <TRANSMISSION02/>
                        <END_OF_CYCLE P0="0"/>
                </CYCLE02>
        </LOOP01>
        <END_OF_LIFE P0="3600"/>
</MISSION>
<TECHNICAL>
        <BUOYANCY_RED P0="1" P1="7" P2="60" P3="7"/>
        <DESCENT P0="30">
                <MIN P0="2.5" P1="1" P2="300" P3="11"/>
                <MAX P0="9" P1="3" P2="100" P3="20"/>
        </DESCENT>
        <DRIFT P0="1800"/>
…
```

4. Technical files

The various pieces of technical information associated with a profile are saved in a binary file organised by sections.

4.1.Transmission of technical information

The system transmits technical files at various stages of the mission:

- During system verification prior to the mission (self-test)
- During the mission at the end of each cycle
- At end of life

4.1.1. Self-test files

The files generated are named using the format "yymmddhhnnss_selfest.hex":

- with "yymmddhhnnss": Date/Time of the float (12 characters)

4.1.2. Mission files

The files generated are named using the format "MxCxTECx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “TECx” : with “x” corresponding the number of the transmission’s session in the cycle

4.1.3. End-of-life files

The various files generated are named using the format "MxCxEOL.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle

4.2. Technical information available

The data is composed of a serial of events. The chronology of these events describes how the cycle unfolded.

An event is defined as follows:

- Type of event (2 Bytes)
- Data event (variable size depending on the type of event)

4.2.1. Product information events

Type number (2 Bytes)	Data event	Format
0	**Product infos**	
	Float ID	CHAR [32]
	Firmware version	
	- Type	CHAR [6]
	- Version	U_INT8
	- Subversion	U_INT8
	- Release	U_INT16
	Firmware checksum	U_INT32
1	**Mission infos**	
	Number of the current mission	U_INT16
	Number of the current cycle	U_INT16
	Cycle's loop setting index	U_INT8
	Cycle setting index	U_INT8
2	**Life infos**	
	Total profile realized	U_INT16
	Total distance profiled (dbar)	IEEE_FLOAT32
	Total duration at sea (s)	U_INT132

4.2.2. Phases events

Type number (2 Bytes)	Data event	Format
1000	**Buoyancy reduction beginning**	
	Date/Time (in second since 2000)	U_INT32
1001	**Buoyancy reduction end**	
	Date/Time (in second since 2000)	U_INT32
	EV lump sum volume (1/10 cm3)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
1002	**Buoyancy inversion beginning**	
	Date/Time (in second since 2000)	U_INT32
1003	**Buoyancy inversion end**	

	Date/Time (in second since 2000)	U_INT32
	External pressure offset (cbar)	U_INT16
	External pressure min (cbar)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
1004	**Descent to parking beginning**	
	Date/Time (in second since 2000)	U_INT32
1005	**Decent to parking end**	
	Date/Time (in second since 2000)	U_INT32
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	External pressure max (cbar)	U_INT16
1006	**Descent to profile beginning**	
	Date/Time (in second since 2000)	U_INT32
1007	**Descent to profile end**	
	Date/Time (in second since 2000)	U_INT32
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	External pressure max (cbar)	U_INT16
1008	**Parking drift beginning**	
	Date/Time (in second since 2000)	U_INT32
1009	**Parking drift end**	
	Date/Time (in second since 2000)	U_INT32
	External pressure min (cbar)	U_INT16
	External pressure max (cbar)	U_INT16
	Stabilized entrance number	U_INT8
	Unstabilized entrance number	U_INT8
	Repositioning number	U_INT8
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8
1010	**Profile drift beginning**	
	Date/Time (in second since 2000)	U_INT32
1011	**Profile drift end**	
	Date/Time (in second since 2000)	U_INT32
	Ext. pressure min (cbar)	U_INT16
	Ext. pressure max (cbar)	U_INT16
	Stabilized entrance number	U_INT8
	Unstabilized entrance number	U_INT8
	Repositioning number	U_INT8
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8



1012	**Ascent beginning**	
	Date/Time (in second since 2000)	U_INT32
1013	**Ascent end**	
	Date/Time (in second since 2000)	U_INT32
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8
1014	**Emergence beginning**	
	Date/Time (in second since 2000)	U_INT32
1015	**In air beginning**	
	Date/Time (in second since 2000)	U_INT32
1016	**Transmission beginning**	
	Date/Time (in second since 2000)	U_INT32
1017	**Emergency ascent beginning**	
	Date/Time (in second since 2000)	U_INT32
	External pressure max (cbar)	U_INT16
1018	**Emergency ascent end**	
	Date/Time (in second since 2000)	U_INT32
	PUMP transferred volume (1/10 cm3)	U_INT16
1019	**End of life**	
	Date/Time (in second since 2000)	U_INT32
	Last date/time of end-of-life detection (in second since 2000)	U_INT32
	End of life source	U_INT32
	- BIT0: Programmed	
	- BIT1: Startup failure	
	- BIT2: Self-test failure	
	- BIT3: Settings failure	
	- BIT4: CTD settings failure	
	- BIT5: Request by downlink	
	- BIT6: Ext. pressure failure	
	- BIT7: Alarm on ext. pressure high	
	- BIT8: Alarm on ext. pressure error	
	- BIT9: Alarm on descent speed high	
	- BIT10: Alarm on battery voltage low	
	- BIT11: Alarm on insulation voltage	
	- BIT12: Alarm on RTC	
1020	**Pressure detection beginning**	
	Date/Time (in second since 2000)	U_INT32
1021	**Pressure detection end**	
	Date/Time (in second since 2000)	U_INT32
	Date/Time of external pressure detection	U_INT32
	External pressure max (dbar)	U_INT16
1022	**Theoretical end of cycle**	
	Date/Time (in second since 2000)	U_INT32



4.2.3. Security events

Type number (2 Bytes)	Data event	Format
2000	**Grounding**	
	Date/Time (in second since 2000)	U_INT32
	External Pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
2001	**Hanging**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
2002	**Braking**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	Speed (mm/s)	U_INT16
2003	**Emergency braking**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	Speed (mm/s)	U_INT16
2004	**ISA detection filtering**	
	Date/Time (in second since 2000)	U_INT32
	Counter of detection	U_INT8
	Number max of detection before confirming ice's presence	U_INT8
2005	**ISA result**	
	Date/Time (in second since 2000)	U_INT32
	Median temperature (m°C)	INT16
2006	**Ice detection**	
	Date/Time (in second since 2000)	U_INT32
	Type of detection	U_INT16
	- BIT0: ISA detection	
	- BIT1: Hanging detection	
	- BIT2: Satellite mask	
	Emergence prohibited until date/time (in second since 2000)	U_INT32
	Emergence will be forced at date/time (in second since 2000)	U_INT32
2007	**Deployment mode**	
	Mode (0: not armed, 1: armed, 2: armed in pressure detection mode)	U_INT8
2008	**New setpoint**	
	Date/Time (in second since 2000)	U_INT32
	Previous setpoint (cbar)	U_INT16
	New setpoint (cbar)	U_INT16
2009	**Empty ballast detection**	
	Date/Time (in second since 2000)	U_INT32

	Detection flag	U_INT8

4.2.4. Alarms events

Type number (2 Bytes)	Data event	Format
3000	**Alarm without value**	
	Date/Time (in second since 2000)	U_INT32
	Alarm type - 11: External pressure sensor →No pressure information - 12: Barometer sensor - 13: Failure on initialization of SBE41 reset offset - 14: Failure on initialization of SBE41 cut-off - 15: Error system - 18: Rescue - 19: GNSS - 20: RTC - 22: DO Aanderaa4330 sensor - 23: Invalid setting of mission - 24: Invalid setting of sensor - 25: Invalid setting of CTD sensor - 26: Empty ballast sensor - 27: Flash memory - 28: SDCard memory - 31: Ice – Satellite mask - 32: Clock drift - 33: CTD SBE41 sensor - 34: CTD RBR sensor - 35: CTD RBR reset offset	U_INT16
3001	**Alarm with value**	
	Date/Time (in second since 2000)	U_INT32
	Alarm type - 0: Low battery voltage - 1: Low external pressure - 2: High external pressure - 3: Low internal pressure - 4: High internal pressure - 5: Low internal temperature - 6: High internal temperature - 7: Low internal humidity - 8: High internal humidity - 9: Gap on external pressure - 10: High speed of descent - 16: Grounding – Volume max reached - 21: Insulation voltage - 29: Hanging – Volume max reached - 30: ISA detection – Number max of detection reached	U_INT16
	Value	IEEE_FLOAT32
	Threshold	IEEE_FLOAT32
3002	**SBE41 limit values**	
	Pressure: Number of samples over limits	U_INT8
	Pressure: Min value expected	IEEE_FLOAT32
	Pressure: Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature: Min value expected	IEEE_FLOAT32

	Temperature: Max value expected	IEEE_FLOAT32
	Salinity: Number of samples over limits	U_INT8
	Salinity: Min value expected	IEEE_FLOAT32
	Salinity: Max value expected	IEEE_FLOAT32
3003	**Aanderaa4330 limit values**	
	C1RPhase: Number of samples over limits	U_INT8
	C1RPhase: Min value expected	IEEE_FLOAT32
	C1RPhase: Max value expected	IEEE_FLOAT32
	C2RPhase: Number of samples over limits	U_INT8
	C2RPhase: Min value expected	IEEE_FLOAT32
	C2RPhase: Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature: Min value expected	IEEE_FLOAT32
	Temperature: Max value expected	IEEE_FLOAT32
3004	**RBRargo limit values**	
	Pressure: Number of samples over limits	U_INT8
	Pressure: Min value expected	IEEE_FLOAT32
	Pressure: Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature: Min value expected	IEEE_FLOAT32
	Temperature: Max value expected	IEEE_FLOAT32
	Salinity: Number of samples over limits	U_INT8
	Salinity: Min value expected	IEEE_FLOAT32
	Salinity: Max value expected	IEEE_FLOAT32
	Coils temperature: Number of samples over limits	U_INT8
	Coils temperature: Min value expected	IEEE_FLOAT32
	Coils temperature: Max value expected	IEEE_FLOAT32
	Salinity corrected: Number of samples over limits	U_INT8
	Salinity corrected: Min value expected	IEEE_FLOAT32
	Salinity corrected: Max value expected	IEEE_FLOAT32
3005	**Sensor error**	
	Number of the sensor	U_INT8
	Number of invalid data	U_INT8
3006	**Invalid frame from sensor**	
	Date/Time (in second since 2000)	U_INT32
	Number of the sensor	U_INT8
	Frame size	U_INT16
	Sensor frame	CHAR [“Frame size”]

4.2.5. Float infos events

Type number (2 Bytes)	Data event	Format

4000	**Battery level**	
	Date/Time (in second since 2000)	U_INT32
	Voltage (mV)	U_INT16
4001	**Battery level at pressure max**	
	Date/Time (in second since 2000)	U_INT32
	Voltage (mV)	U_INT16
	External pressure (cbar)	U_INT16
4002	**Servitude measurement**	
	Date/Time (in second since 2000)	U_INT32
	Internal pressure (mbar)	U_INT16
	Internal temperature (m°C)	INT16 + offset (20000m°C)
	Internal relative humidity (%RH)	U_INT8
4003	**Internal pressure compensated**	
	Internal pressure (mbar)	U_INT16
4004	**Reset**	
	Date/Time (in second since 2000)	U_INT32
	Source of the reset	U_INT8
	- BIT0 : Firewall	
	- BIT1 : Option byte loader	
	- BIT2 : PIN reset	
	- BIT3 : BOR reset	
	- BIT4 : software	
	- BIT5 : Independent watchdog	
	- BIT6 : Window watchdog	
	- BIT7 : Low-power reset	
4005	**Memory disk**	
	Disk number (0: Flash, 1: SDcard)	U_INT8
	Disk occupancy (resolution: 0.5%)	U_INT8
4006	**Last loss of internal vacuum**	
	Date/Time (in second since 2000)	U_INT32
	Internal pressure (mbar)	U_INT16
	Internal temperature (m°C)	INT16 + offset (20000m°C)
	Internal relative humidity (%RH)	U_INT8

4.2.6. Sensors state events

Type number (2 Bytes)	Data event	Format
5000	**SBE41 subsurface point**	
	CTD pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
5001	**SBE41 offset pressure**	
	Offset value	IEEE_FLOAT32
5002	**RBRargo3 offset pressure**	

	Offset value	IEEE_FLOAT32

4.2.7. Selftest events

Type number (2 Bytes)	Data event	Format
6000	**Self-test result**	
	FRAM memory result flag (0: error, 1: ok)	U_INT8
	FLASH memory result flag (0: error, 1: ok)	U_INT8
	SDcard state (0: error, 1: ok, 2: not present)	U_INT8
	XML settings result flag (0: error, 1: ok)	U_INT8
	RTC result flag (0: error, 1: ok)	U_INT8
	RTC date/Time (in second since 2000)	U_INT32
	Insulation result flag (0: error, 1: ok)	U_INT8
	Insulation value	U_INT16
	Battery result flag (0: error, 1: ok)	U_INT8
	On load battery voltage (mV)	U_INT16
	EV result flag (0: error, 1: ok)	U_INT8
	PUMP result flag (0: error, 1: ok)	U_INT8
	Empty ballast sensor result flag (0: error, 1: ok)	U_INT8
	Max buoyancy pumping duration (s)	U_INT16
	Internal pressure result flag (0: error, 1: ok)	U_INT8
	Internal pressure (mbar)	IEEE_FLOAT32
	Internal pressure compensed (mbar)	IEEE_FLOAT32
	Internal temperature result flag (0: error, 1: ok)	U_INT8
	Internal temperature (°C)	IEEE_FLOAT32
	Internal relative humidity result flag (0: error, 1: ok)	U_INT8
	Internal relative humidity (%)	IEEE_FLOAT32
	GNSS result flag (0: error, 1: ok)	U_INT8
	Modem result flag (0: error, 1: ok)	U_INT8
	Dummy buffer (allowing future evolutions)	U_INT8 * 32
6001	**SBE41 self-test result**	
	Fast pressure result flag (0: error, 1: ok)	U_INT8
	Fast pressure (dbar)	IEEE_FLOAT32
	CTD measurement result flag (0: error, 1: ok)	U_INT8
	CTD Pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
6002	**Aanderaa4330 self-test result**	
	Measurement result flag (0: error, 1: ok)	U_INT8
	Serial number	U_INT16
6003	**RBRargo3 self-test result**	
	Fast pressure result flag (0: error, 1: ok)	U_INT8
	Fast pressure (dbar)	IEEE_FLOAT32
	CTD measurement result flag (0: error, 1: ok)	U_INT8

	CTD pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
	CTD coils temperature (°C)	IEEE_FLOAT32
	CTD corrected salinity (PSU)	IEEE_FLOAT32
	Sensor's serial number	U_INT32
6100	**FRAM memory self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6101	**FLASH memory self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6102	**SDcard memory self-test result**	
	State (0: error, 1: ok, 2: not present)	U_INT8
6103	**XML settings self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6104	**RTC self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
	RTC date/Time (in second since 2000)	U_INT32
6105	**Insulation measurement self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
	Value	U_INT16
6106	**Battery self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
	On load battery voltage (mV)	U_INT16
6107	**EV self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6108	**PUMP self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6109	**Empty ballast sensor self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
	Max buoyancy pumping duration (s)	U_INT16
6110	**Internal measurement self-test result**	
	Internal pressure result flag (0: error, 1: ok)	U_INT8
	Internal pressure (mbar)	IEEE_FLOAT32
	Internal pressure compensated (mbar)	IEEE_FLOAT32
	Internal temperature result flag (0: error, 1: ok)	U_INT8
	Internal temperature (°C)	IEEE_FLOAT32
	Internal relative humidity result flag (0: error, 1: ok)	U_INT8
	Internal relative humidity (%)	IEEE_FLOAT32
6111	**GNSS self-test result**	
	Result flag (0: error, 1: ok)	U_INT8
6112	**Modem self-test result**	
	Result flag (0: error, 1: ok)	U_INT8

4.2.8. Transmission events

Type number (2 Bytes)	Data event	Format
7000	**Geographic location**	
	Date/Time (in second since 2000)	U_INT32
	Latitude (°)	IEEE_FLOAT32
	Longitude (°)	IEEE_FLOAT32
	Number of satellites	U_INT8
	Valid fix	U_INT8
	Clock drift (s) **(*)**	IEEE_FLOAT32
	Session duration (s)	U_INT16
7001	**SBD Iridium (*)**	
	Date/Time of session's end (in second since 2000)	U_INT32
	Session duration (s)	U_INT16
	SBDI success counter	U_INT16
	SBDI total counter	U_INT16
	Remote command accepted counter	U_INT8
	Remote command refused counter	U_INT8
	Remote command unknown counter	U_INT8
	Received files number	U_INT16
	Received size (Byte)	IEEE_FLOAT32
	Transmitted files number	U_INT16
	Transmitted size (Byte)	IEEE_FLOAT32
	Files number in waiting of transmission	U_INT16
7002	**RUDICS Iridium (*)**	
	Date/Time of session's end (in second since 2000)	U_INT32
	Session duration (s)	U_INT16
	Session number	U_INT8
	Remote command accepted counter	U_INT8
	Remote command refused counter	U_INT8
	Remote command unkown counter	U_INT8
	Received files number	U_INT16
	Received size (Byte)	IEEE_FLOAT32
	Transmitted files number	U_INT16
	Transmitted size (Byte)	IEEE_FLOAT32
	Files number in waiting of transmission	U_INT16

4.2.9. Navigation events

Type number (2 Bytes)	Data event	Format
8000	**First stabilization (on descent)**	
	Date/Time (in second since 2000)	U_INT32
	Stabilization state (0x00: not stabilized, 0x01: stabilized)	U_INT8
	External pressure (cbar)	U_INT16
8001	**Pressure monitoring**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
8002	**Pump action**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
8003	**Ev action**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16

4.2.10. Treatment events

Type number (2 Bytes)	Data event		Format
9000	**Treated data in profile**		
	Date/Time (in second since 2000)		U_INT32
	Number of the sensor		U_INT8
	Number of the phase		U_INT8
	Zone 1	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 2	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 3	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 4	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 5	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
9001	**Treated data in drift**		
	Date/Time (in second since 2000)		U_INT32
	Number of the sensor		U_INT8
	Number of the phase		U_INT8
	Number of samples		U_INT16



(*) Clock drift = Float time - GPS time. The offset is reset to 0 after every GPS retiming.

(*) Information from the previous transmission session.

Example of parsing:

```
Previous Iridium session infos ->   2023/12/11 02:58:29
                                    Session duration :      462 s
                                    SBDI success :          21 / 21
                                    Command accepted :      0
                                    Command refused :       0
                                    Command unknown :       0
                                    SBD received :          0 (Size: 0 B)
                                    SBD transmitted :       5 (Size: 6645 B)
                                    Files in waiting :      0
Product life -> Total profiles realized: 255 Total distance profiled: 34997 dbar Total duration at sea: 35 days, 5 hours
### MISSION 95 - CYCLE 21 (LOOP01 - CYCLE01) ###
### BUOYANCY REDUCTION - 2023/12/11 02:58:32 ###
Internal measures ->    Pressure:       1009 mbar
                        Temperature:    20.158 °C
                        Humidity:       51 %RH
Battery voltage:        10.669 V
SBE41 - Pressure offset:    0.000 cbar
Ev forfaitaire volume:      0.0 cm3
Ev transfered volume:       686.1 cm3
Ev action number:           99
### BUOYANCY REDUCTION END - 2023/12/11 03:11:24 ###

### DESCENT TO PARKING - 2023/12/11 03:11:25 ###
Pressure infos -> 2023/12/11 03:16:30 Pressure: 22.4 dbar
Pressure infos -> 2023/12/11 03:21:30 Pressure: 33.2 dbar
Pressure infos -> 2023/12/11 03:26:32 Pressure: 41.2 dbar
Pressure infos -> 2023/12/11 03:31:32 Pressure: 47.3 dbar
Pressure infos -> 2023/12/11 03:36:32 Pressure: 51.7 dbar
Pressure infos -> 2023/12/11 03:41:32 Pressure: 55.1 dbar
1st stabilization -> 2023/12/11 03:41:32 Pressure: 55.1 dbar
Ev action -> 2023/12/11 03:41:33 Pressure: 55.1 dbar Transfered volume: 0.3 cm3
Pressure infos -> 2023/12/11 03:46:32 Pressure: 57.9 dbar
Ev action -> 2023/12/11 03:46:34 Pressure: 57.9 dbar Transfered volume: 11.0 cm3
Pressure infos -> 2023/12/11 03:51:32 Pressure: 74.0 dbar
Pressure infos -> 2023/12/11 03:56:32 Pressure: 86.1 dbar
Pressure infos -> 2023/12/11 04:01:32 Pressure: 95.2 dbar
Pressure infos -> 2023/12/11 04:06:34 Pressure: 102.0 dbar
Pressure infos -> 2023/12/11 04:11:34 Pressure: 107.0 dbar
Pressure infos -> 2023/12/11 04:16:34 Pressure: 110.8 dbar
Pressure infos -> 2023/12/11 04:21:34 Pressure: 113.6 dbar
Ev action -> 2023/12/11 04:21:36 Pressure: 113.6 dbar Transfered volume: 11.0 cm3
Pressure infos -> 2023/12/11 04:26:34 Pressure: 133.9 dbar
Pressure infos -> 2023/12/11 04:31:34 Pressure: 149.0 dbar
Pressure infos -> 2023/12/11 04:36:34 Pressure: 160.4 dbar
Pressure infos -> 2023/12/11 04:41:36 Pressure: 168.9 dbar
Pressure infos -> 2023/12/11 04:46:36 Pressure: 175.2 dbar
Pressure infos -> 2023/12/11 04:51:36 Pressure: 179.9 dbar
Pressure infos -> 2023/12/11 04:56:36 Pressure: 183.5 dbar
Pressure infos -> 2023/12/11 05:01:36 Pressure: 186.1 dbar
Ev action -> 2023/12/11 05:01:37 Pressure: 186.1 dbar Transfered volume: 6.3 cm3
Pressure infos -> 2023/12/11 05:06:36 Pressure: 204.7 dbar
Pressure infos -> 2023/12/11 05:11:36 Pressure: 218.7 dbar
Pressure infos -> 2023/12/11 05:16:36 Pressure: 229.2 dbar
```



Pressure infos -> 2023/12/11 05:21:38 Pressure: 237.0 dbar
Pressure infos -> 2023/12/11 05:26:38 Pressure: 242.8 dbar
Pressure infos -> 2023/12/11 05:31:38 Pressure: 247.2 dbar
Pressure infos -> 2023/12/11 05:36:38 Pressure: 250.4 dbar
Ev action -> 2023/12/11 05:36:39 Pressure: 250.4 dbar Transfered volume: 2.1 cm3
Pressure infos -> 2023/12/11 05:41:40 Pressure: 260.0 dbar
Pressure infos -> 2023/12/11 05:46:40 Pressure: 267.1 dbar
Pressure infos -> 2023/12/11 05:51:40 Pressure: 272.4 dbar
Pressure infos -> 2023/12/11 05:56:40 Pressure: 276.3 dbar
Pressure infos -> 2023/12/11 06:01:42 Pressure: 279.3 dbar
Ev transfered volume: 30.7 cm3
Ev action number: 5
Pressure max: 279.3 dbar
DESCENT TO PARKING END - 2023/12/11 06:01:43

PARKING DRIFT - 2023/12/11 06:01:43
Treated data infos -> Sensor: 1 Phase: DESC_TO_PARK Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
Zone 2: Sample number= 190 Treatment= AVERAGE DECIMATED
Zone 3: Sample number= 80 Treatment= AVERAGE DECIMATED
Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Treated data infos -> Sensor: 2 Phase: DESC_TO_PARK Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
Zone 2: Sample number= 240 Treatment= AVERAGE DECIMATED
Zone 3: Sample number= 30 Treatment= AVERAGE DECIMATED
Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Pressure min: 279.4 dbar
Pressure max: 279.4 dbar
Stabilized entrance number: 0
Unstabilized entrance number: 0
Repositioning number: 0
Ev transfered volume: 0.0 cm3
Ev action number: 0
Pump transfered vol.: 0.0 cm3
Pump action number: 0
PARKING DRIFT END - 2023/12/11 06:01:54

DESCENT TO PROFILE - 2023/12/11 06:01:55
Ev transfered volume: 0.0 cm3
Ev action number: 0
Pressure max: 279.5 dbar
DESCENT TO PROFILE END - 2023/12/11 06:02:05

PROFILE DRIFT - 2023/12/11 06:02:05
Pressure min: 279.6 dbar
Pressure max: 279.6 dbar
Stabilized entrance number: 0
Unstabilized entrance number: 0
Repositioning number: 0
Ev transfered volume: 0.0 cm3
Ev action number: 0
Pump transfered vol.: 0.0 cm3
Pump action number: 0
PROFILE DRIFT END - 2023/12/11 06:02:14

ASCENT - 2023/12/11 06:02:15
Internal measures -> Pressure: 1010 mbar
Temperature: 20.748 °C
Humidity: 51 %RH
Pump action -> 2023/12/11 06:02:34 Pressure: 279.7 dbar Transfered volume: 11.0 cm3
Battery voltage min: 9.498 V (279.7 dbar)
Pump action -> 2023/12/11 06:04:38 Pressure: 269.4 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:16:38 Pressure: 186.1 dbar Transfered volume: 11.0 cm3



```
Pump action -> 2023/12/11 06:24:38 Pressure: 139.5 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:28:38 Pressure: 118.4 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:38:38 Pressure: 57.1 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:44:40 Pressure: 22.1 dbar Transfered volume: 11.0 cm3
Pump transfered vol.:                77.0 cm3
Pump action number:                  7
### ASCENT END - 2023/12/11 06:46:31 ###

### EMERGENCE - 2023/12/11 06:46:35 ###
Pump action -> 2023/12/11 06:51:59 Pressure: 0.0 dbar Transfered volume: 420.0 cm3
### TRANSMISSION - 2023/12/11 06:52:04 ###
Memory disk usage -> Drive: FLASH    Space used: 26.5 %
Memory disk usage -> Drive: SDCARD   Space used: 0.0 %
Subsurface point ->          Pressure:          5.192 dbar
                             Temperature:       0.000 °C
                             Salinity:          32.680 PSU
Treated data infos -> Sensor: 1 Phase: ASCENT   Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
                                                Zone 2: Sample number= 190 Treatment= AVERAGE DECIMATED
                                                Zone 3: Sample number= 80 Treatment= AVERAGE DECIMATED
                                                Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
                                                Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Treated data infos -> Sensor: 2 Phase: ASCENT   Zone 1: Sample number= 7 Treatment= AVERAGE DECIMATED
                                                Zone 2: Sample number= 224 Treatment= AVERAGE DECIMATED
                                                Zone 3: Sample number= 24 Treatment= AVERAGE DECIMATED
                                                Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
                                                Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Geographic location ->       UTC :              2023/12/11 06:53:05
                             Latitude :         47.7978287 °
                             Longitude :        -3.2856851 °
                             Satellite number:  3
                             Valid fix :        1
                             Clock drift :      0.424 s
                             Session duration:  57 s
```

5. Sensor data files

The system generates sensor data files in different formats depending on the sensors.

5.1.Sensors identification

Each file embedded the number of the sensor as “Sx”, where “x” is the number of the sensor.
Each sensor are identified as :

- **S1**: SBE41
- **S2**: Aanderaa 4330
- **S3**: RBRargo3

5.2.Files format

5.2.1. Iridium SBD format

The files generated are named using the format "MxCxSxFxYx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “Sx” : with “x” corresponding the number of the sensor (see 5.1.)
- “Fx” : with “x” corresponding the type of data’s format (different for each sensor)
- “Yx” : with “Y” corresponding the phase
  - “D”: Descent to parking
  - “P”: Parking drift
  - “T”: Descent to profile
  - “B”: Profile drift
  - “A”: Ascent
  - “I”: In air

  And with “x” being the dating frequency where:
  - 0: No date
  - 1: One date on each sensor’s data
  - 15: One date every 15 sensor’s data

5.2.2. Iridium RUDICS format

The files generated are named using the format "MxCxSxFx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “Sx” : with “x” corresponding the number of the sensor (see 5.1.)
- “Fx” : with “x” corresponding the type of data’s format (different for each sensor)

Unlike the SBD format, each sample is dated.

5.2.2.1. Phase information format

The phase informations are encoded by setting the date to 0xFFFFFFF.

Value	Format	Calculation
Date/Time	U_INT32	Set to 0xFFFFFFFF
Phase	U_INT8	- 0: Descent to parking - 1: Parking drift - 2: Descent to profile - 3: Profile drift - 4: Ascent - 5: Surface
Zone of acquisition	U_INT8	From 1 to 5
Treatment type	U_INT8	- 0x00 : raw - 0x80 : raw decimated - 0x81 : average decimated

The values corresponding to the acquisition zone and the type of treatment are set to 0xFF for the phases “Parking drit”, “Profile drift” and “Surface” phases.

5.3. Data format

5.3.1. SBE41 - “S1”

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
F1	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n
F2	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	20	P(dbar) = n/20 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000

5.3.2. Aanderaa 4330 - “S2”

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	C1RPhase (°)	U_INT16	40	500	C1(°) = n/500 – 40
	C2RPhase (°)	U_INT16	40	500	C2(°) = n/500 – 40
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
F1	Date/Time	U_INT32			Dating depends on file

	(in second since 2000)				name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	C1RPhase (°)	IEEE_FLOAT32			C1(°) = n
	C2RPhase (°)	IEEE_FLOAT32			C2(°) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
F2	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	20	P(dbar) = n/20 - 100
	C1RPhase (°)	U_INT16	40	500	C1(°) = n/500 – 40
	C2RPhase (°)	U_INT16	40	500	C2(°) = n/500 – 40
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10

5.3.3. RBRargo3 – "S3"

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
F1	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Coils temperature (°C)	U_INT16	10	1000	Tc(°C) = n/1000 - 10
F2	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F3	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Coils temperature (°C)	U_INT16	10	1000	Tc(°C) = n/1000 - 10
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F4	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F5	Date/Time (in second since 2000)	U_INT32			Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n

	Temperature (°C)	IEEE_FLOAT32	T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32	S(PSU) = n
F6	Date/Time (in second since 2000)	IEEE_FLOAT32	Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32	P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32	T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32	S(PSU) = n
	Coils temperature (°C)	IEEE_FLOAT32	Tc(°C) = n
F7	Date/Time (in second since 2000)	IEEE_FLOAT32	Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32	P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32	T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32	S(PSU) = n
	Salinity corrected (PSU)	IEEE_FLOAT32	Sc(PSU) = n
F8	Date/Time (in second since 2000)	IEEE_FLOAT32	Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32	P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32	T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32	S(PSU) = n
	Coils temperature (°C)	IEEE_FLOAT32	Tc(°C) = n
	Salinity corrected (PSU)	IEEE_FLOAT32	Sc(PSU) = n
F9	Date/Time (in second since 2000)	IEEE_FLOAT32	Dating depends on file name
	Pressure (dbar)	IEEE_FLOAT32	P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32	T(°C) = n
	Salinity corrected (PSU)	IEEE_FLOAT32	Sc(PSU) = n

Example: M95C7S1F0A15.hex (Mission: 95, Cycle: 7, Sensor: SBE41, Data format: F0, Phase: ascent, dating frequency: 1/15)

M95C7S1F0A15.hex

000002ad	00 01	02 03	04 05	06 07	08 09	0a 0b	0c 0d	0e 0f
00000000	92 41	06 2d	84 0f	0a 37	b0 86	7d 0f	0a 37	b0 86
00000010	73 0f	0a 37	b0 86	69 0f	0a 37	b0 86	5f 0f	05 37
00000020	b0 86	55 0f	00 37	b0 86	4b 0f	00 37	ab 86	40 0f
00000030	00 37	a6 86	37 0f	00 37	a6 86	2e 0f	00 37	a6 86
00000040	23 0f	00 37	a6 86	19 0f	fc 36	a6 86	10 0f	f6 36
00000050	a6 86	05 0f	f6 36	a6 86	fb 0e	f6 36	a6 86	3d 42
00000060	06 2d	f1 0e	f6 36	a1 86	e7 0e	f6 36	9c 86	dd 0e
00000070	f1 36	9c 86	d3 0e	e7 36	9c 86	c9 0e	e2 36	9c 86
00000080	bf 0e	e2 36	9c 86	b5 0e	e2 36	97 86	aa 0e	dc 36
00000090	92 86	a1 0e	d3 36	92 86	98 0e	ce 36	92 86	8e 0e
000000a0	ca 36	92 86	83 0e	c4 36	92 86	79 0e	c4 36	92 86
000000b0	6f 0e	c4 36	92 86	65 0e	c4 36	8c 86	c5 42	06 2d
000000c0	5a 0e	c4 36	88 86	50 0e	be 36	88 86	46 0e	ba 36
000000d0	88 86	3c 0e	ba 36	88 86	33 0e	b5 36	88 86	29 0e
000000e0	b0 36	88 86	1f 0e	ab 36	83 86	14 0e	a6 36	7e 86
000000f0	0b 0e	ab 36	7e 86	01 0e	b0 36	7e 86	f6 0d	b6 36
00000100	7e 86	ed 0d	ba 36	7e 86	e4 0d	ba 36	7e 86	d9 0d
00000110	ba 36	7e 86	cf 0d	ba 36	7e 86	33 43	06 2d	c4 0d
00000120	ba 36	7e 86	ba 0d	c0 36	7e 86	b1 0d	c4 36	7e 86
00000130	a8 0d	c4 36	7e 86	9e 0d	c4 36	7e 86	94 0d	c4 36
00000140	7a 86	8a 0d	c0 36	74 86	7f 0d	b5 36	74 86	74 0d
00000150	a4 36	6e 86	6a 0d	96 36	6a 86	61 0d	8d 36	6a 86
00000160	57 0d	83 36	6a 86	4e 0d	7e 36	6a 86	43 0d	7e 36
00000170	65 86	39 0d	73 36	60 86	ad 43	06 2d	2f 0d	65 36
00000180	60 86	25 0d	5b 36	5b 86	1b 0d	51 36	56 86	11 0d
00000190	4c 36	56 86	07 0d	51 36	56 86	fd 0c	56 36	56 86
000001a0	f3 0c	56 36	56 86	e9 0c	5b 36	56 86	de 0c	5a 36
000001b0	50 86	d5 0c	56 36	4c 86	cb 0c	56 36	4c 86	c1 0c
000001c0	51 36	4c 86	b7 0c	47 36	4c 86	ac 0c	42 36	46 86
000001d0	a2 0c	42 36	42 86	3a 44	06 2d	98 0c	3c 36	42 86
000001e0	8e 0c	32 36	42 86	84 0c	1d 36	3c 86	7a 0c	10 36
000001f0	38 86	71 0c	10 36	38 86	67 0c	10 36	38 86	5e 0c
00000200	10 36	38 86	53 0c	10 36	38 86	49 0c	10 36	38 86
00000210	3f 0c	1a 36	38 86	35 0c	1f 36	33 86	2b 0c	1a 36

Date

CTD sample

```
M95C7S1F0A15.hex
SBE41 - ASCENT
Measure Date/Time      P(dbar)     Temp(°C)    Sal.(PSU)
2023/12/08 21:05:22     297.20000     4.09000    34.48000
                        296.50000     4.09000    34.48000
                        295.50000     4.09000    34.48000
                        294.50000     4.09000    34.48000
                        293.50000     4.08500    34.48000
                        292.50000     4.08000    34.48000
                        291.50000     4.08000    34.47500
                        290.40000     4.08000    34.47000
                        289.50000     4.08000    34.47000
                        288.60000     4.08000    34.47000
                        287.50000     4.08000    34.47000
                        286.50000     4.07600    34.47000
                        285.60000     4.07000    34.47000
                        284.50000     4.07000    34.47000
                        283.50000     4.07000    34.47000
2023/12/08 21:08:13     282.50000     4.07000    34.46500
                        281.50000     4.07000    34.46000
                        280.50000     4.06500    34.46000
                        279.50000     4.05500    34.46000
                        278.50000     4.05000    34.46000
                        277.50000     4.05000    34.46000
                        276.50000     4.05000    34.45500
                        275.40000     4.04400    34.45000
                        274.50000     4.03500    34.45000
                        273.60000     4.03000    34.45000
                        272.60000     4.02600    34.45000
                        271.50000     4.02000    34.45000
                        270.50000     4.02000    34.45000
                        269.50000     4.02000    34.45000
                        268.50000     4.02000    34.44400
2023/12/08 21:10:29     267.40000     4.02000    34.44000
                        266.40000     4.01400    34.44000
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